ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

Kenan Song
School of Chips
Email:
Kenan.Song@xjtlu.edu.cn
Office: C5007

Coding for Structure Building and Band Structure

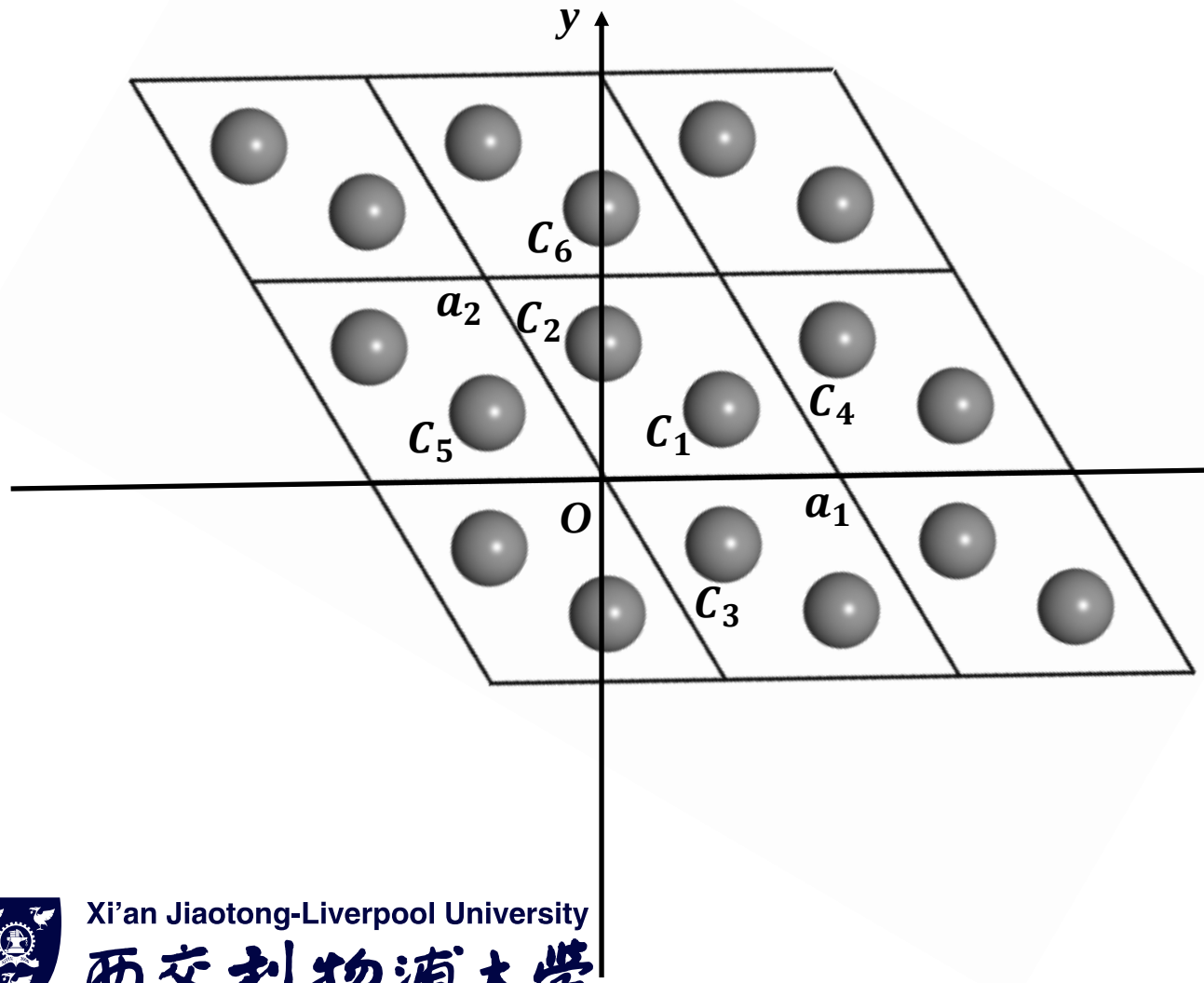
I. Tight Binding Hamiltonian for Pristine Graphene

II. Coding for TBM of Graphene

III. Tuning band structure of Graphene from Conductor into a Semiconductor

I. Tight Binding Hamiltonian for Pristine Graphene

Structure of Graphene



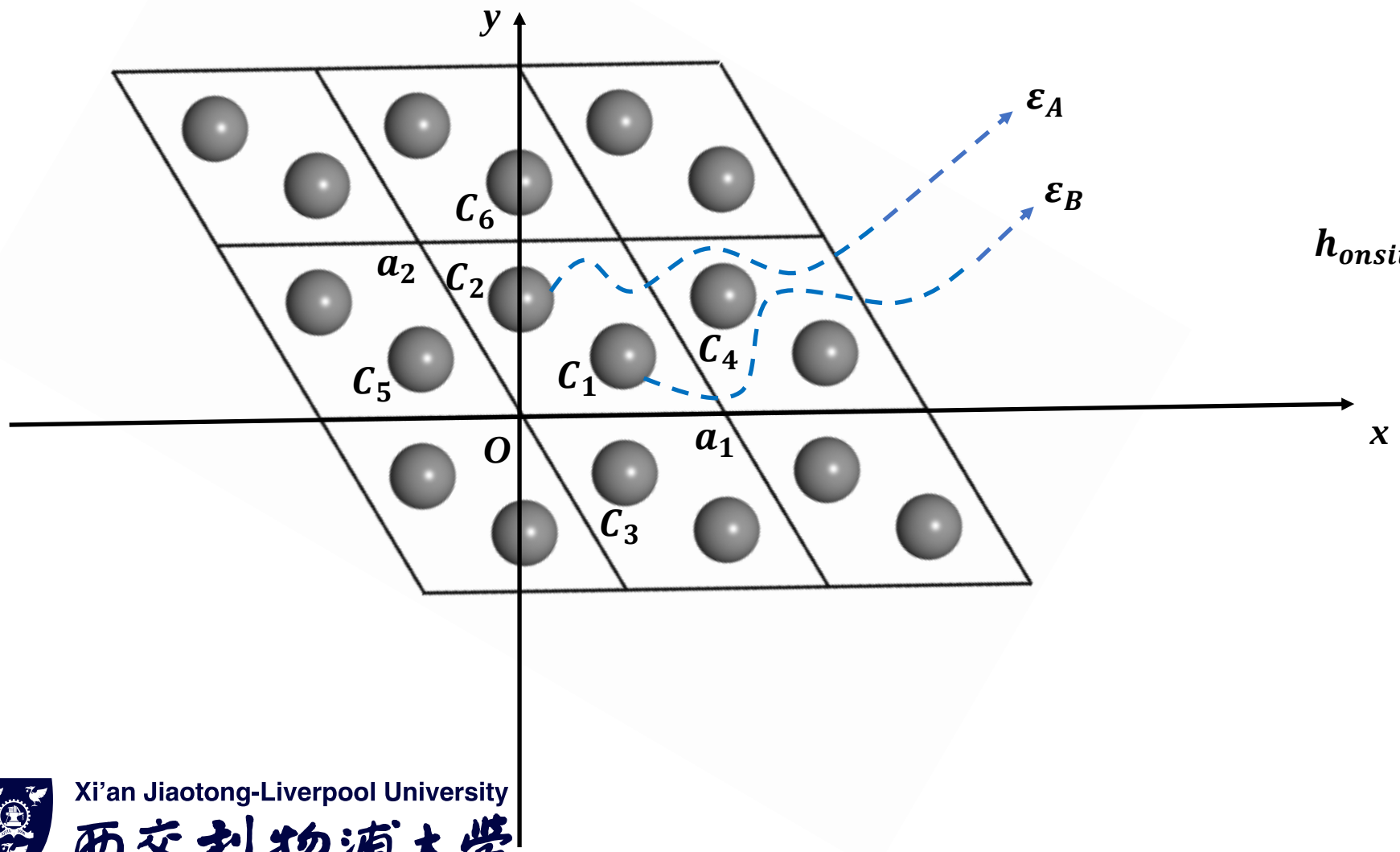
$$|a_1| = |a_2| = a = 2.46\ \text{Å}$$

$$a_1 = \begin{pmatrix} a \\ 0.0 \\ 0.0 \end{pmatrix}; a_2 = \begin{pmatrix} a \times cos120^\circ \\ a \times sin120^\circ \\ 0.0 \end{pmatrix}$$

$$C_1 = \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \\ 0.0 \end{pmatrix}; C_2 = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ 0.0 \end{pmatrix}$$

I. Tight Binding Hamiltonian for Pristine Graphene

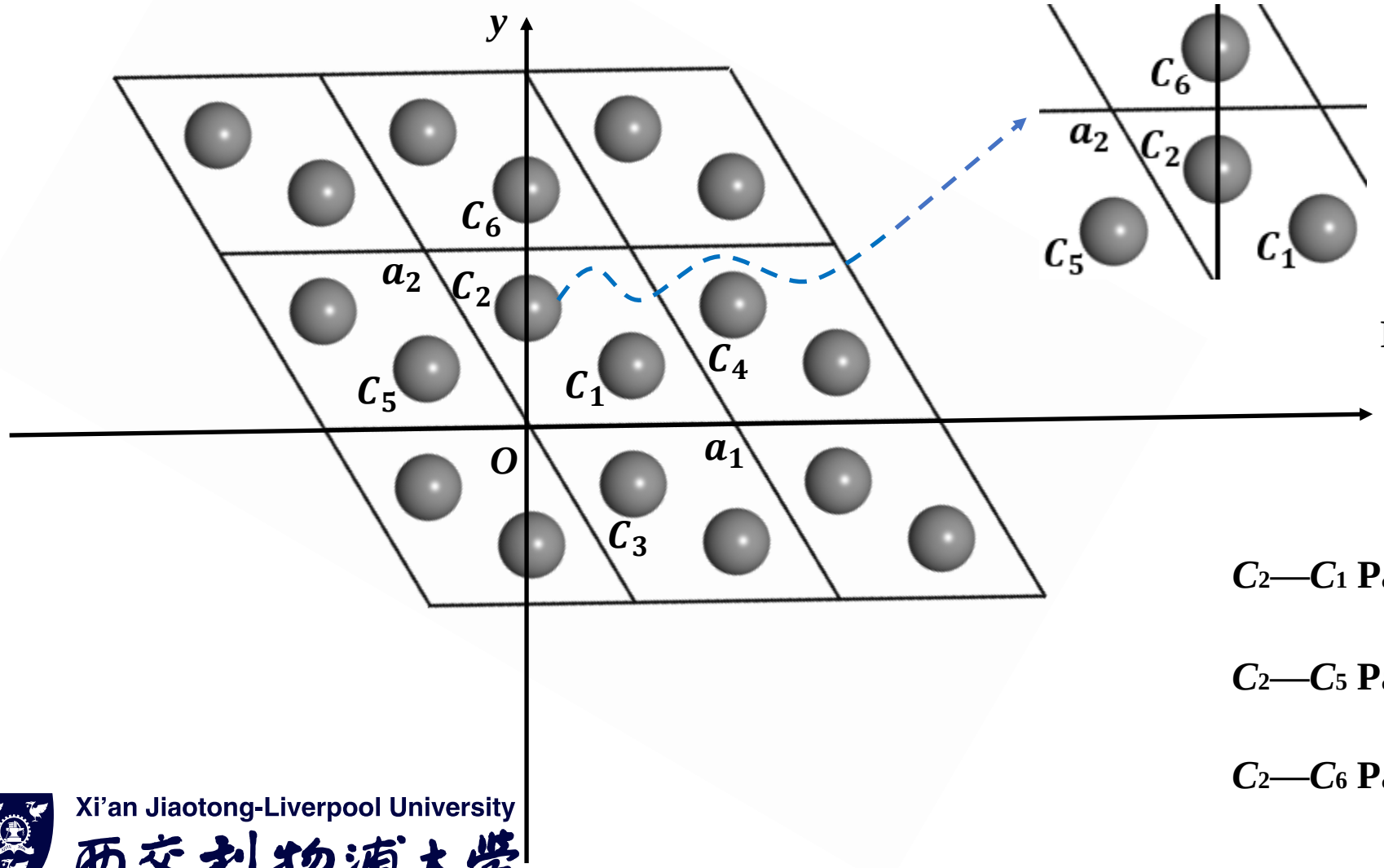
Onsite Energy for Graphene



$$h_{onsite} = \begin{vmatrix} \varepsilon_A & 0 \\ 0 & \varepsilon_B \end{vmatrix}$$

I. Tight Binding Hamiltonian for Pristine Graphene

First Neighbour Hopping term for Graphene



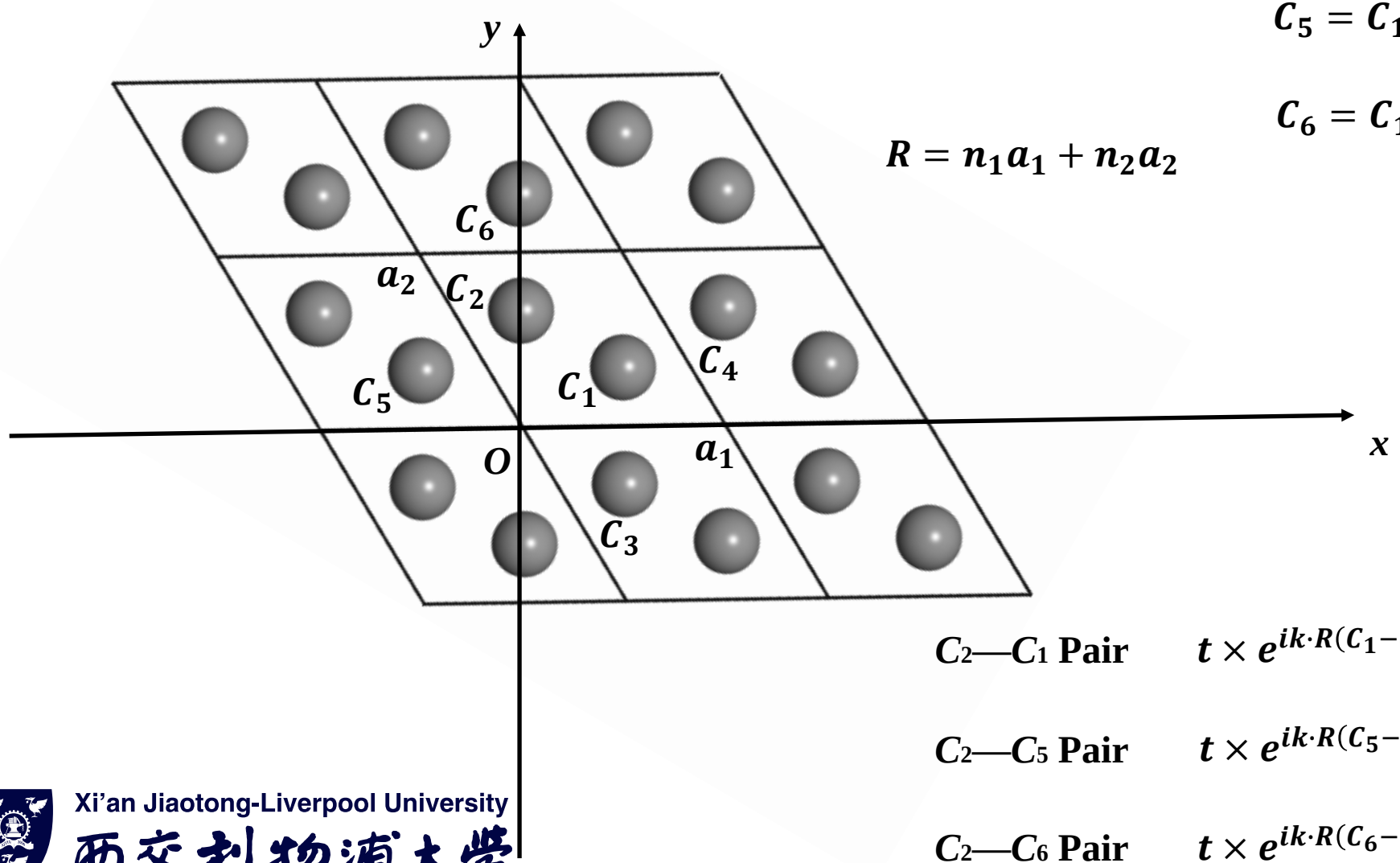
First Neighbour for C_2: C_1, C_5 and C_6

C_2—C_1 Pair $t \times e^{ik \cdot R(C_1 - C_2)}$

C_2—C_5 Pair $t \times e^{ik \cdot R(C_5 - C_2)}$

C_2—C_6 Pair $t \times e^{ik \cdot R(C_6 - C_2)}$

I. Tight Binding Hamiltonian for Pristine Graphene



$$C_5 = C_1 + (-1 \times a_1 + 0 \times a_2)$$

$$C_6 = C_1 + (0 \times a_1 + 1 \times a_2)$$

$$R = n_1 a_1 + n_2 a_2$$

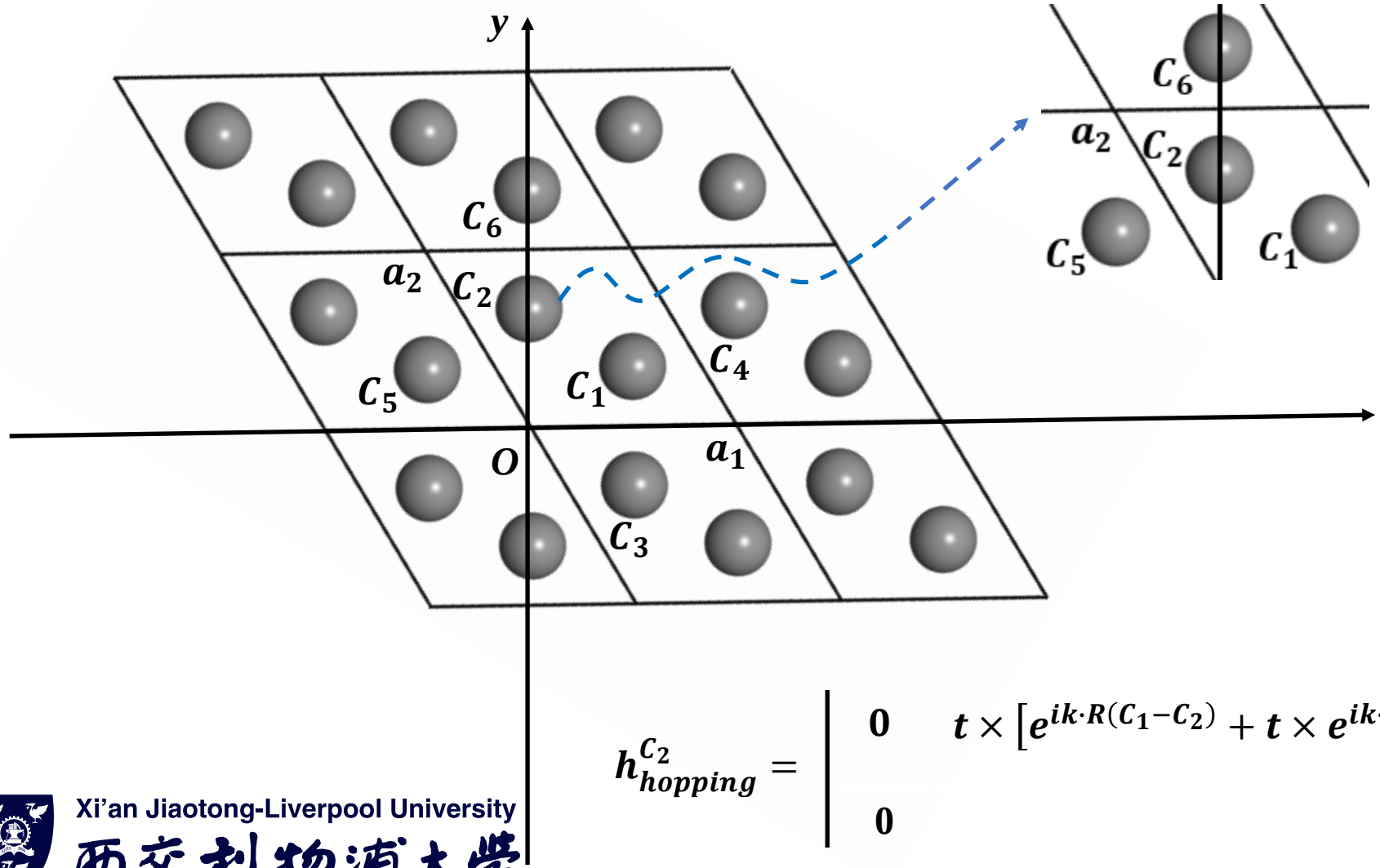
C_2—C_1 Pair $\quad t \times e^{ik \cdot R(C_1 - C_2)}$

C_2—C_5 Pair $\quad t \times e^{ik \cdot R(C_5 - C_2)} = t \times e^{ik \cdot R(C_1 - a_1 - C_2)}$

C_2—C_6 Pair $\quad t \times e^{ik \cdot R(C_6 - C_2)} = t \times e^{ik \cdot R(C_1 + a_2 - C_2)}$

I. Tight Binding Hamiltonian for Pristine Graphene

First Neighbour Hopping term for Graphene

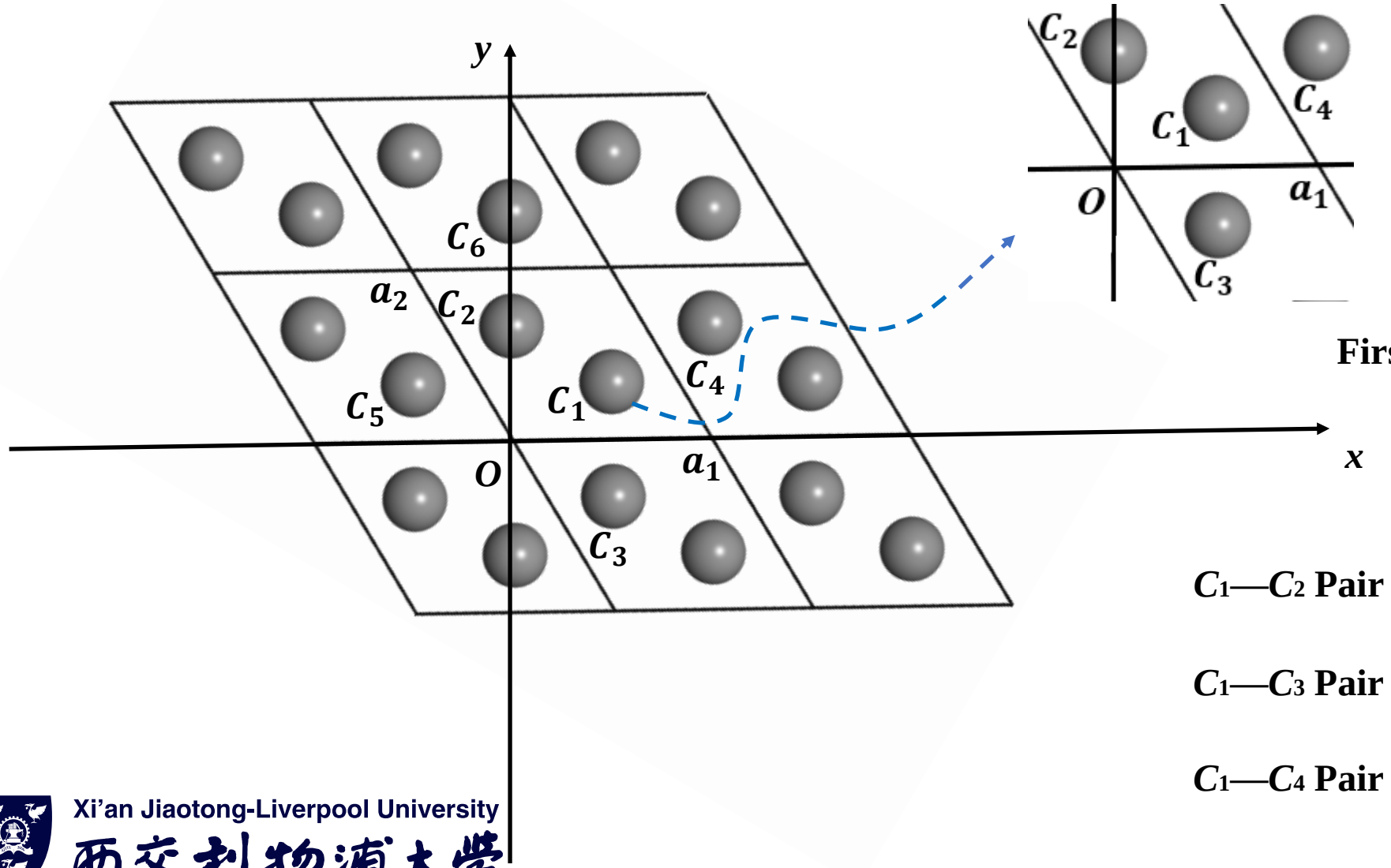


First Neighbour for C_2: C_1, C_5 and C_6

$$h_{hopping}^{C_2} = \begin{vmatrix} 0 & t \times \left[e^{ik \cdot R(C_1 - C_2)} + t \times e^{ik \cdot R(C_1 - a_1 - C_2)} + t \times e^{ik \cdot R(C_1 + a_2 - C_2)} \right] \\ 0 & 0 \end{vmatrix}$$

I. Tight Binding Hamiltonian for Pristine Graphene

First Neighbour Hopping term for Graphene



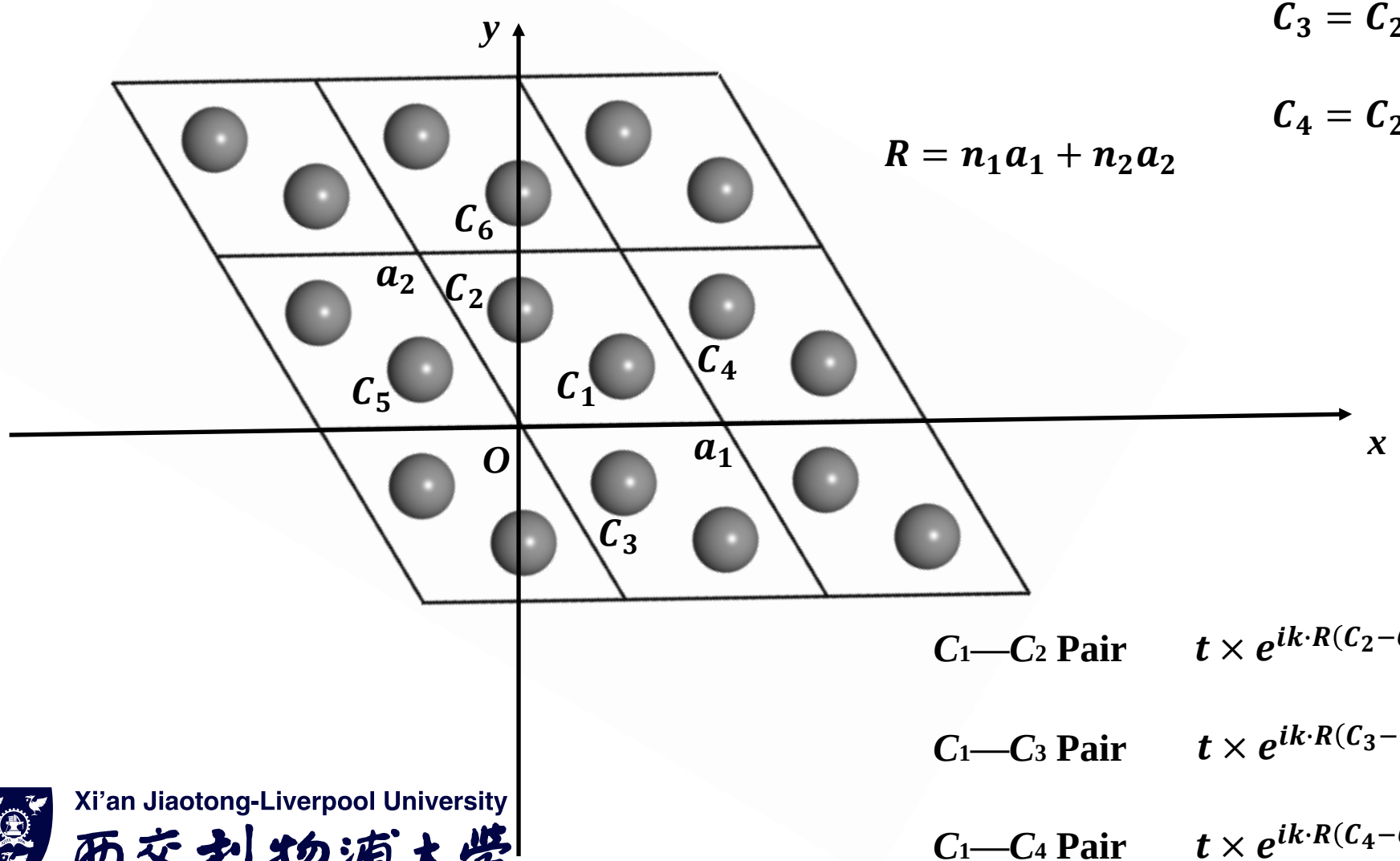
First Neighbour for C_1: C_2, C_3 and C_4

C_1—C_2 Pair $t \times e^{ik \cdot R(C_2 - C_1)}$

C_1—C_3 Pair $t \times e^{ik \cdot R(C_3 - C_1)}$

C_1—C_4 Pair $t \times e^{ik \cdot R(C_4 - C_1)}$

I. Tight Binding Hamiltonian for Pristine Graphene



$$R = n_1 a_1 + n_2 a_2$$

$$C_3 = C_2 + (0 \times a_1 - 1 \times a_2)$$

$$C_4 = C_2 + (1 \times a_1 + 0 \times a_2)$$

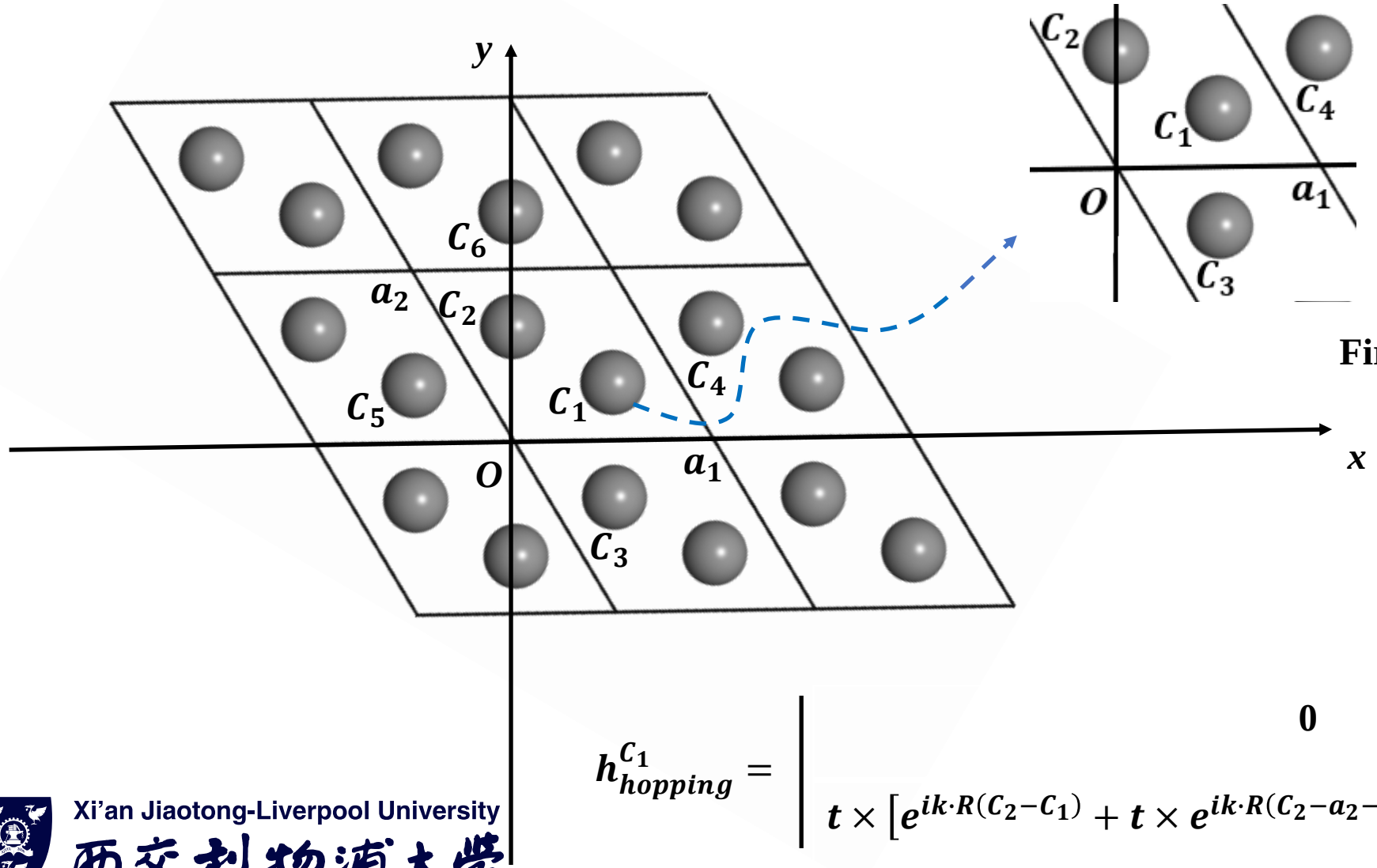
C_1—C_2 Pair $\quad t \times e^{ik \cdot R(C_2 - C_1)}$

C_1—C_3 Pair $\quad t \times e^{ik \cdot R(C_3 - C_1)} = t \times e^{ik \cdot R(C_2 - a_2 - C_1)}$

C_1—C_4 Pair $\quad t \times e^{ik \cdot R(C_4 - C_1)} = t \times e^{ik \cdot R(C_2 + a_1 - C_1)}$

I. Tight Binding Hamiltonian for Pristine Graphene

First Neighbour Hopping term for Graphene



First Neighbour for C_1: C_2, C_3 and C_4

$$h_{hopping}^{C_1} = \begin{vmatrix} 0 & 0 \\ t \times \left[e^{ik\cdot R(C_2-C_1)} + t \times e^{ik\cdot R(C_2-a_2-C_1)} + t \times e^{ik\cdot R(C_2+a_1-C_1)}\right] & 0 \end{vmatrix}$$

I. Tight Binding Hamiltonian for Pristine Graphene

$$h_{hopping}^{C_2} = \begin{vmatrix} 0 & t \times \left[e^{ik \cdot R(C_1 - C_2)} + t \times e^{ik \cdot R(C_1 - a_1 - C_2)} + t \times e^{ik \cdot R(C_1 + a_2 - C_2)} \right] \\ 0 & \end{vmatrix}$$

Conjugate Transpose

$$h_{hopping}^{C_1} = \begin{vmatrix} & 0 \\ t \times \left[e^{ik \cdot R(C_2 - C_1)} + t \times e^{ik \cdot R(C_2 - a_2 - C_1)} + t \times e^{ik \cdot R(C_2 + a_1 - C_1)} \right] & 0 \end{vmatrix}$$

$$h_{hopping} = \begin{vmatrix} 0 & t \times \left[e^{ik \cdot R(C_1 - C_2)} + t \times e^{ik \cdot R(C_1 - a_1 - C_2)} + t \times e^{ik \cdot R(C_1 + a_2 - C_2)} \right] \\ h.c. & 0 \end{vmatrix}$$

I. Tight Binding Hamiltonian for Pristine Graphene

$$hamiltonian = h_{onsite} + h_{hopping} = \begin{vmatrix} \varepsilon_A & t \times \left[e^{ik\cdot R(C_1-C_2)} + t \times e^{ik\cdot R(C_1-a_1-C_2)} + t \times e^{ik\cdot R(C_1+a_2-C_2)}\right] \\ h.c. & \varepsilon_B \end{vmatrix}$$

$$e^{ik\cdot R(C_1-C_2)} = cos[k \cdot R(C_1 - C_2)] + i \times sin[k \cdot R(C_1 - C_2)]$$

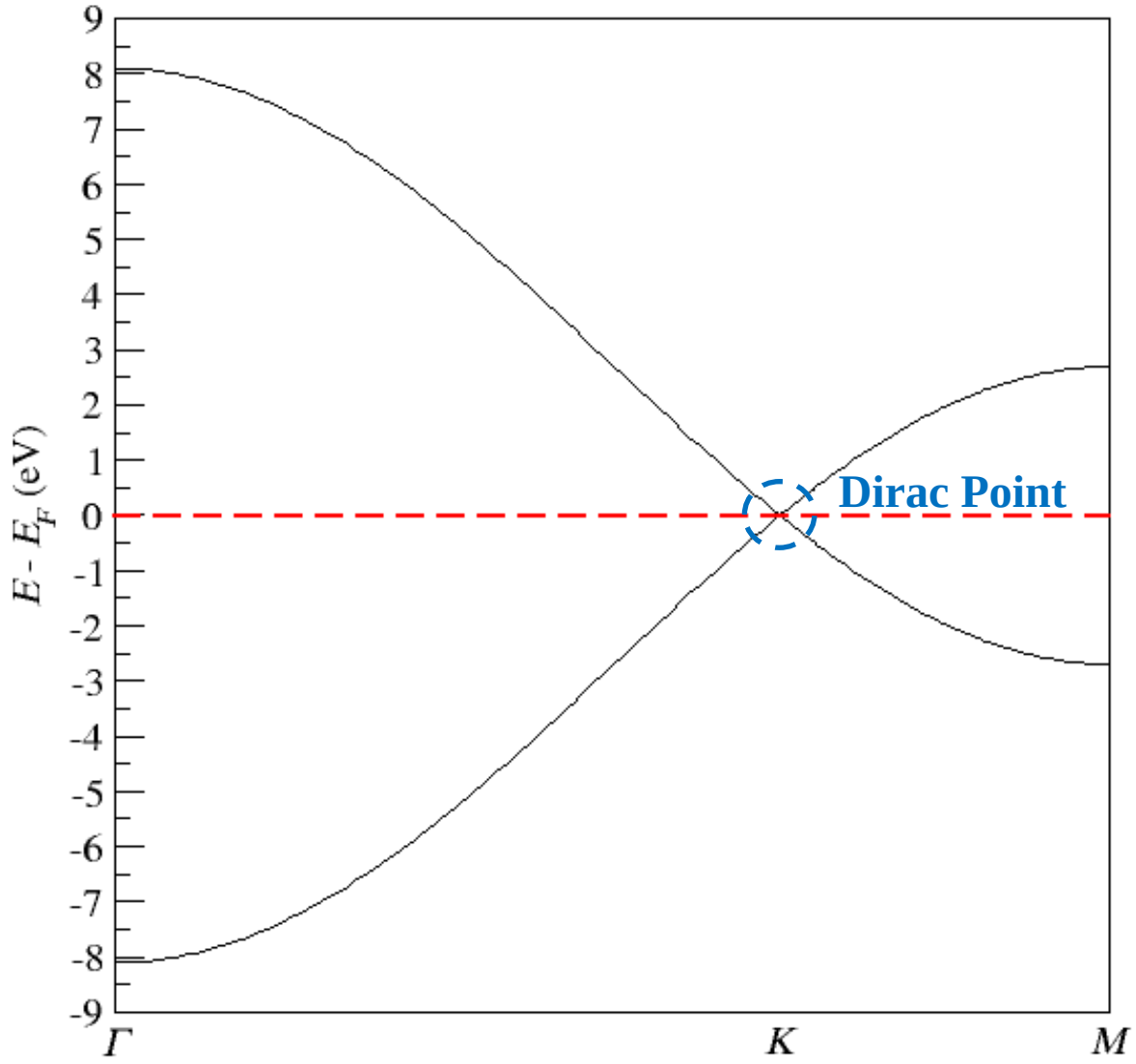
$$k \cdot R(C_1 - C_2) = k_x \times R_x + k_y \times R_y$$

Complex-type Matrix

Check the Fortran Code

II. Coding for TBM of Graphene

Band Structure for the Pristine Graphene



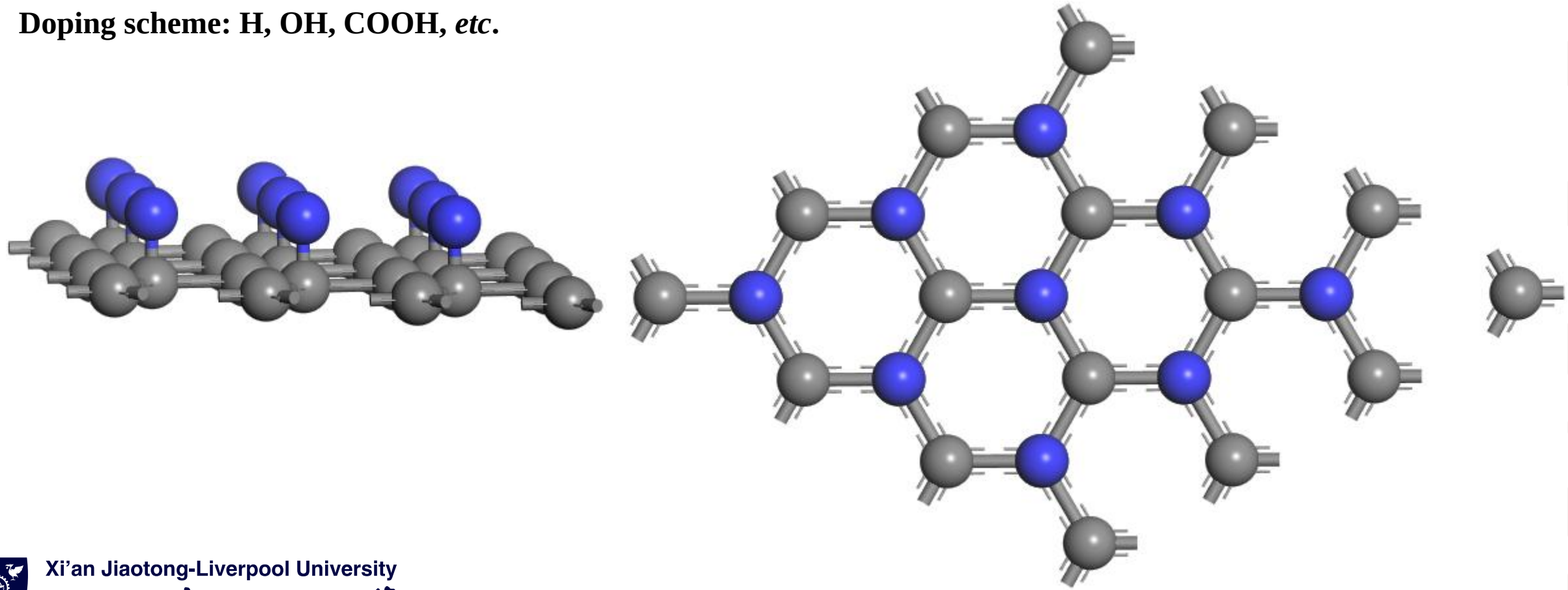
The Position of Dirac Point is controlled by the onsite energy

If $\varepsilon_A = \varepsilon_B = \varepsilon \neq 0$, DP shows up at ε eV

III. Tuning band structure of Graphene from Conductor into a Semiconductor

Create different onsite energy for these two sublattices

Doping scheme: H, OH, COOH, *etc*.



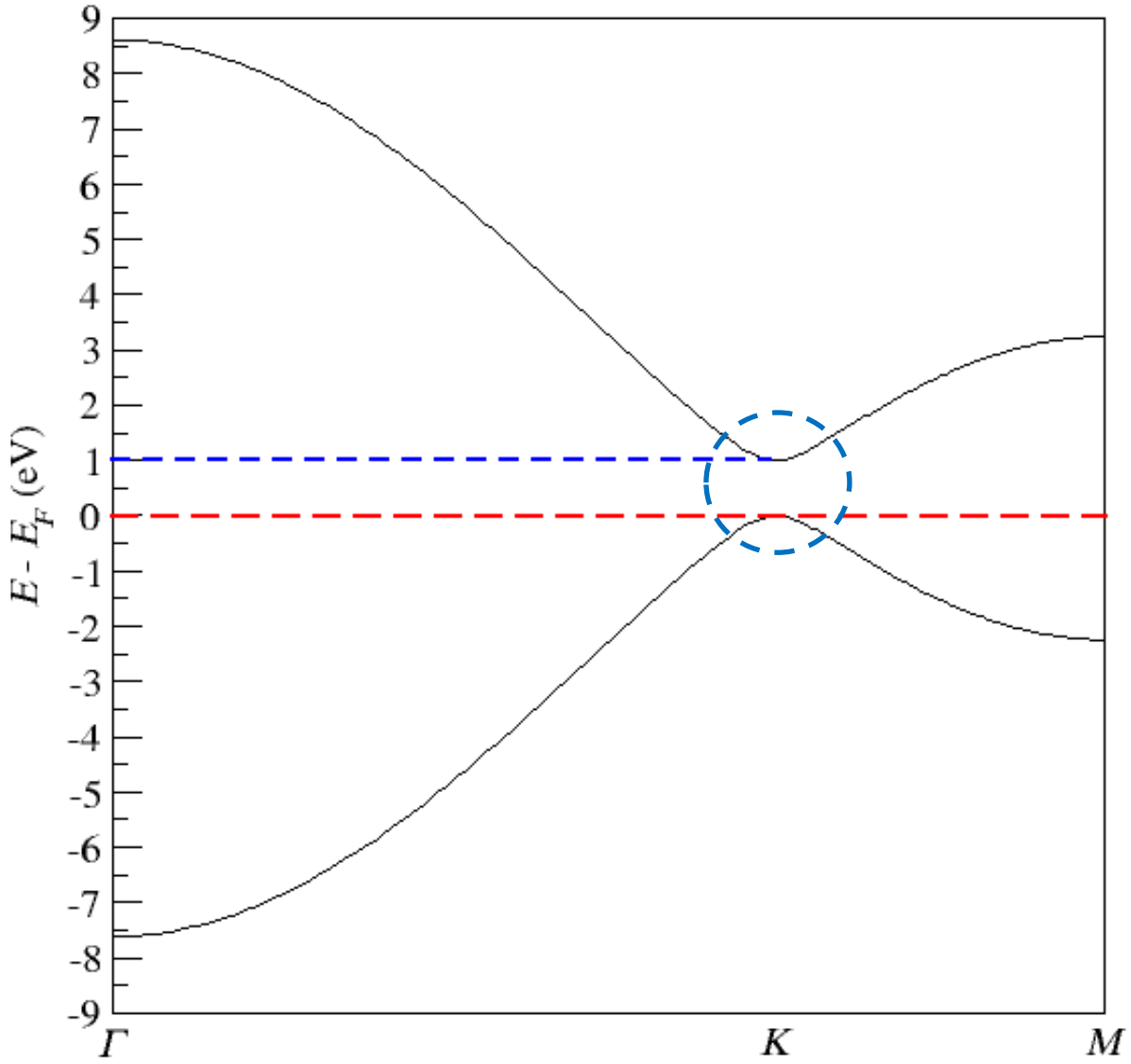
III. Tuning band structure of Graphene from Conductor into a Semiconductor

$$hamiltonian = h_{onsite} + h_{hopping} = \begin{vmatrix} \varepsilon_A & t \times \left[e^{ik\cdot R(C_1 - C_2)} + t \times e^{ik\cdot R(C_1 - a_1 - C_2)} + t \times e^{ik\cdot R(C_1 + a_2 - C_2)}\right] \\ h.c. & \varepsilon_B \end{vmatrix}$$

Make $\varepsilon_A \neq \varepsilon_B$; for instance, $\varepsilon_A = 1\ eV$ and $\varepsilon_B = 0\ eV$

III. Tuning band structure of Graphene from Conductor into a Semiconductor

Band Structure for the doped Graphene



Dirac Point is gone and a gap shows up

Gap = $| \varepsilon_A - \varepsilon_B |$

III. Tuning band structure of Graphene from Conductor into a Semiconductor

In practice, doping is a universal strategy for tuning the band gap

Depending on the oxidizing or reducing nature of the dopants, the gap can be modulated to either a larger or smaller value



That's all for this session

Thank you for your attention and great participation